How to Build a Bird Kite



by Laurel Schwulst

This is the printable companion to
the video tutorial “How to Build a Bird Kite,”
published May 19, 2021 in the New York Times

https://www.nytimes.com/interactive/2021/05/19/style/how-to-build-a-bird-kite.html

Materials



plastic

A generic medium-grade plastic will do, technically called 1 mil or 2 ply. Personally, I like using mattress bags for their transparent look — but feel free to experiment with a different kind or color, like painters tarp, garbage bags, plastic tablecloth, etc. — as long as it's roughly medium grade and doesn't have holes.



wooden dowels

Look for long, thin wooden dowels at your local hardware store. You'll need 4 dowels for this design, but I'd get at least 5 (in case one breaks). I bought 1/8" diameter, 48" long wooden dowels.



electrical tape

Any electrical tape will do, as it's weather resistant and elastic. (Most regular tape is brittle and becomes unusable if it gets wet.) Feel free to use nice bright colored electrical tape, maybe some that looks good with your plastic.



kite string

Also called "flying line" — technically, any lightweight, long string will do. But this item may also be purchased online or at a specialty store. I searched for 30-50lb nylon or Dacron flying line, which usually comes on a nice spool for easy flying. Here's one I like on Amazon: https://amzn.to/3euaOCO



scissors

Any scissors that can cut plastic & thin wooden dowels will do



permanent marker

So that you can draw the bird shape on the plastic

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Instructions

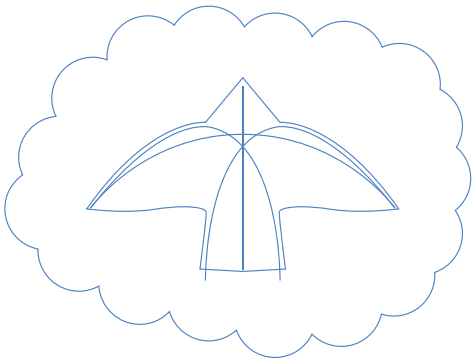
1. Get ready

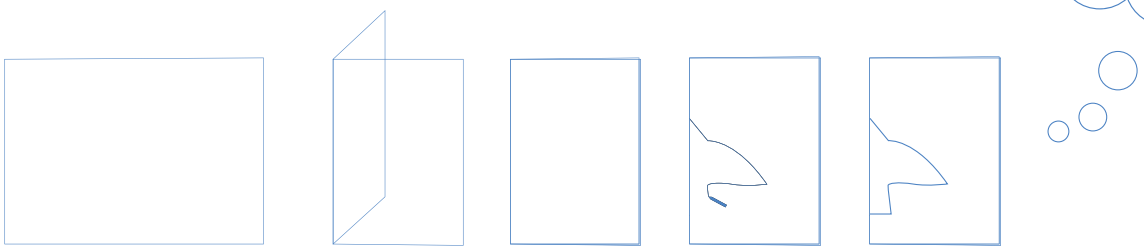
Gather your materials — from a hardware store, from online, etc.

Set up your working area — clear a nice area on the floor, free of debris, with all your materials at the ready. Pour yourself a glass of water and put on some white noise or calm background music. The creation of your working environment is just as important as the creation of the kite itself.

2. Fold plastic in half

Make sure it's completely smooth. You should roughly have a 3 ft x 3 ft area once it's folded. You should have one edge that's the folded edge. This is the part we are working with. In a few steps, you will draw your design along this folded edge — which might remind you of making snowflakes out of paper.





3. Draw half the bird shape on the plastic

First, test the bend of your dowels. There's a minimum bendable radius any dowel can handle. Try feeling out this limit without breaking your dowel.

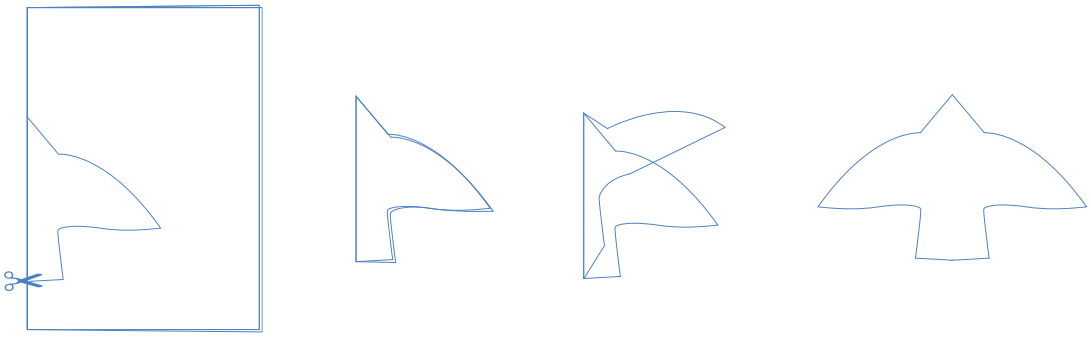
Next, envision how your dowels will be placed on the bird (using the drawing in the cloud shape above). One dowel will go straight down the middle. The curve of the other dowels will become the curve of your wings.

Next, you'll draw the bird shape. Remember, because we are using the folded edge of the plastic, we are only drawing half of the bird shape. Here is how to do it: Place a dowel on the plastic and bend it in an arc. Use your permanent marker to draw along the curve of the dowel. This is the wing. At the top of this curve, add a triangle. The triangle is the head. To complete the half bird shape, add an angled rectangle at the bottom. This is the tail.

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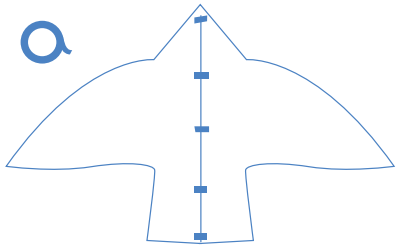
4. Cut out the bird shape from the plastic

Using your scissors, cut out the bird from the plastic.
Then unfold it so you have your bird shape complete!

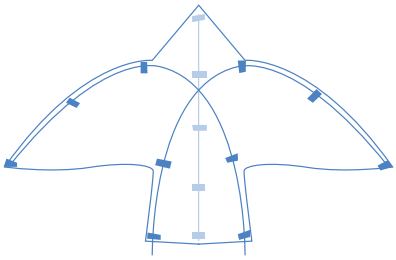


5. Apply your dowels to the bird

You'll apply four dowels:

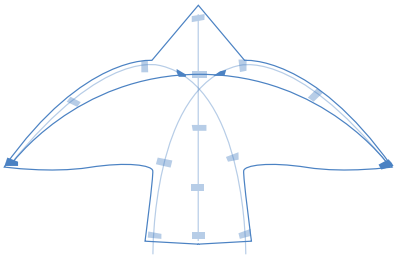


#1 — The first is the "vertical spar" down the center of the bird. You may likely need to cut the end of the dowel to make it fit nicely. Once it's the right size and in the right position (completely centered), tape it on both ends and in between a few times.



#2 & #3 — The next two dowels, the "spreader spars," will be symmetrical and curved. They will start at the wing tip and end at the opposite side's tail feather. Tape both ends strongly and in between a few times.

Also, it's fine if your dowel extends past your plastic near the tail feathers a little bit. This could be a cool stylistic choice.

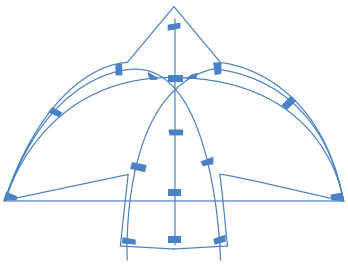


#4 — The final, fourth dowel is technically optional. But it makes your bird look very cool. This dowel goes from wingtip to wingtip. You can tape it at each tip again and where it intersects the spreader spars.

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6. Tie a string from wingtip to wingtip

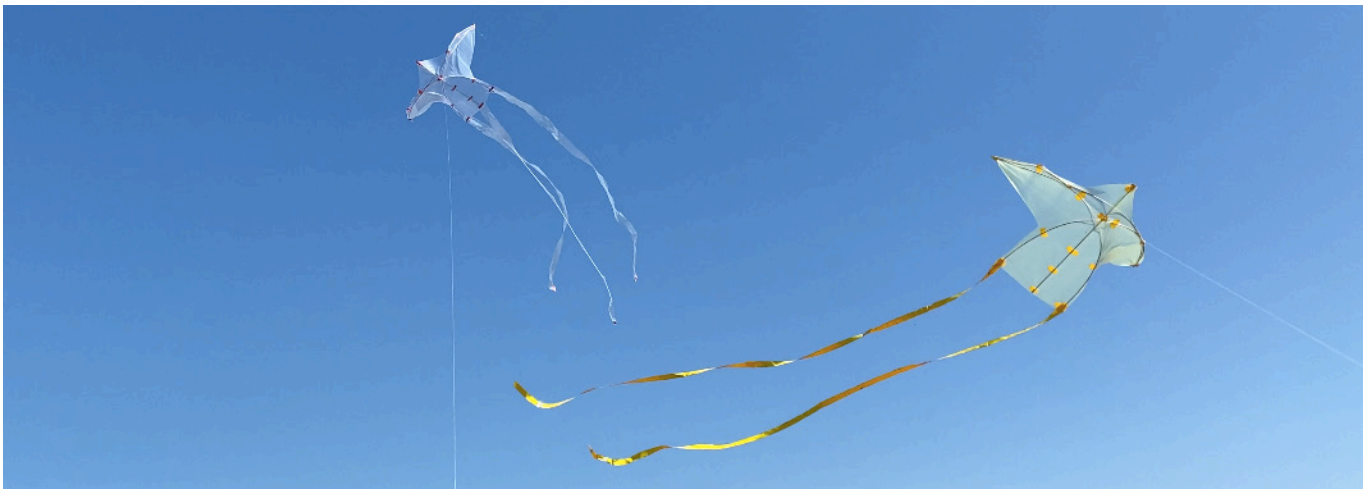
To make your kite have 3D shape, cut off some of your flying string and tie each end to a wingtip. This will give your bird some volumetric form in the air.



7. Add the bird's extended tail

Using spare plastic, create some long narrow strips. They should be at least 4 times the length of the kite itself and about 2 inches wide each.

Using tape, attach your strips of plastic to the end of your kite near the bird's tail feathers. If you like, add even more strips for a beautiful effect — I liked adding three tails, one taped to each dowel. Note that this tail will be functional, and not just stylish. It helps to weigh the kite down to help it fly well.



8. Get ready to fly

In preparation for flying, gather all your materials (scissors, tape, string, extra dowels, etc.) and your bird kite — maybe in a large bag for easy transport. You'll want to bring these materials with you because you might need to make some adjustments once you actually fly your kite.

Now that your kite is constructed, it's time for your last ingredient: the wind. Choose an open area that feels often or consistently windy… a beach is usually a fine option. Parks can work too but winds are less guaranteed. I recommend researching online wind forecasts and scheduling your flight on a promisingly windy day.

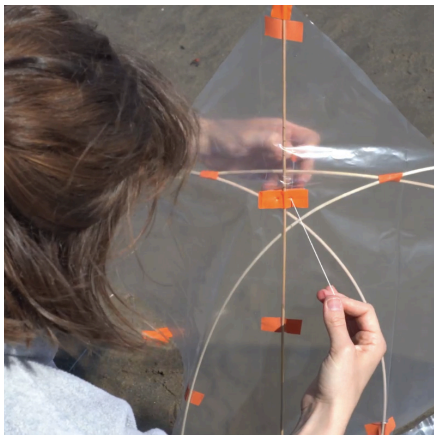
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9. Apply kite string to the bird

At your flying location, first hold up your bird kite and feel which way it wants to fly. Then imagine where you might best attach your kite string. It is likely near the central heart of the bird, around where all the spars meet — somewhere along the central spar that feels most stable and centered. Hold the kite from this point.

Tape the end of your kite string to your kite securely at the central point you have just identified. With your back to the wind, slowly release the bird and see if it flies.



If that went well, remove the tape and use string to attach the line more securely. To do this, poke two holes in your plastic with your scissors and thread the end of the kite string through, around the vertical spar, and back. (To make the plastic more resilient, you might consider applying electrical tape to the area before poking your holes.) Once you have your string through, make a secure knot. (If you're feeling pro, do a "square knot" so you know it's secure.)



10. Try flying your bird kite

Now, stand with your back to the wind, and let the flying line out as your bird rises into the sky…

Remember that wind is key. Winds change day to day. You might need to come back on a day in which there's more wind. Or, if the wind isn't strong, try running with your kite so it has a chance to get up into the air — most kites need some sort of running start in weak winds. Or maybe the dimensions of your kite need a different string placement. Try some other things out with your bag of materials at the ready. Be willing to experiment and modify your bird kite, "kite surgery," on the fly.

Have fun flying!

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